



KENYA INSTITUTE OF CURRICULUM DEVELOPMENT

Nurturing Every Learner's Potential

SENIOR SCHOOL CURRICULUM DESIGN

GRADE 10

MATHEMATICS

MARCH, 2025

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FOREWORD

The Government of Kenya is committed to ensuring that policy objectives for Education, Training and Research meet the provisions of the Kenya Constitution 2010 and Vision 2030, East African Community (EAC) Protocol, African Agenda 2063, United Nations Sustainable Development Goals (SDGs) and demands of the 21st Century.

Towards achieving the vision and mission of Basic Education, the Ministry of Education (MoE) has successfully and progressively rolled out the Competency Based Curriculum (CBC) at Pre-Primary (PP1& PP2), Primary (Grades 1-6) and Junior School (Grades 7-9) levels of education. The first cohort of CBC learners will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level facilitate learners to acquire relevant knowledge, skills, attitudes and values in preparation for Tertiary Education and the world of work.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

I urge all Government agencies and other stakeholders in Education to use the designs to plan for effective and efficient implementation of the Competency Based Curriculum.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Curriculum (CBC) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The CBC was piloted in 2017 and 2018 and rolled out at Pre-Primary, Primary and Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through a phase in - phase out strategy, to ensure a smooth transition. Currently, the first Cohort of CBC are at Grade 9 – the last grade at Junior School. The implementation of CBC has also benefited immensely from the recommendations of the Presidential Working Party on Education Reforms (PWPER) that were released in August 2023 and fully implemented, in terms of Curriculum design and curriculum support materials.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways and tracks of their choice, guided by their potentials and interests as demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and/or development of initial competencies for the Industry. This is an important step in the realisation of the Vision of the curriculum reforms; which is to produce *engaged, empowered and ethical citizens*. The vision and mission of CBC as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs indicate, among others, suggested learning experiences and resources. The Ministry of Education, through the State Department for Basic Education commits to provide these resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers and other officers involved in supporting and quality assuring curriculum implementation to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 mandates the Institute to develop curricula and curriculum support materials for Basic and Tertiary Education. The curriculum development process involves thorough research, and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD, 2017).

KICD receives its funding from the Government of Kenya (GOK) to enable the successful achievement of its mandate. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank through the Kenya Primary Education Equity in Learning Programme (KPEELP); a project coordinated by MoE. Therefore, the Institute is very grateful to the GOK, through the MoE and the development partners for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also acknowledge the KICD staff, all teachers, educators who took part as panelists and various stakeholders. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide implementation of the CBC at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. Foster nationalism and patriotism and promote national unity.

Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. Promote the social, economic, technological and industrial needs for national development.

Education should prepare the youth of the country to play an effective and productive role in the life of the nation. **a) Social Needs**

Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.

b) Economic Needs

Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building a modern and independent economy which is in need of an adequate and relevant domestic workforce.

c) Technological and Industrial Needs

Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. Promote individual development and self-fulfilment

Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

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4. Promote sound moral and religious values.

Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. Promote social equity and responsibility.

Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. Promote respect for and development of Kenya's rich and varied cultures.

Education should instil in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

7. Promote international consciousness and foster positive attitudes towards other nations.

Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. Promote positive attitudes towards good health and environmental protection.

Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to

physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

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THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners will transition to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre university/ pre-career experience in which the learners have an opportunity to choose pathways and tracks where they shall have demonstrated interest and potential at the earlier levels. Senior School comprises three years of education for learners in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be *engaged, empowered and ethical citizens* ready to participate in the socio economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise four core learning areas namely: English, Kiswahili, Community Service Learning (CSL) and Physical Education (PE). The learner will select three learning areas depending on career choice, aptitude, interest and personality and with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. Communicate effectively and utilise information and communication technology across varied contexts.

2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems.
4. Exploit individual talents for leisure, self-fulfilment, career growth, further education and training.
5. Uphold national, moral and religious values and apply them in day-to-day life.
6. Apply and promote health care strategies in day-to-day life.
7. Protect, preserve and improve the environment for sustainability.
8. Demonstrate active local and global citizenship for harmonious co-existence.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

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LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
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1. English 2. Kiswahili/KSL 3. Community Service Learning (CSL) 4. Physical Education (PE)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Mathematics 24. Biology 25. Chemistry 26. Physics 27. General Science 28. Agriculture 29. Computer Studies 30. Home Science 31. Aviation 32. Building Construction 33. Electricity 34. Metalwork 35. Power Mechanics 36. Woodwork 37. Media Technology 38. Marine and Fisheries Technology
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LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	

1. English	6
2. Kiswahili	6
3. Physical Education (PE)	3
4. Community Service Learning (CSL)	3
Elective Subjects	
5. Elective 1	6
6. Elective 2	6
7. Elective 3	6
<i>* ICT Skills</i>	2
<i>*PPI</i>	1
<i>* Learner Personal /Group Study</i>	1
Total Number of Lessons	40

***Note:**

a) ICT skills will be offered to all learners to facilitate learning and enjoyment. These skills include accessing, creating, digital citizenship, and caring for ICT devices. The teacher should use ICT skills for delivery of lessons. b) Every school

shall offer Pastoral Programme Instruction (PPI) for moral, spiritual and character development of the learners.

ESSENCE STATEMENT

In Senior School, Mathematics builds on and deepens the competencies acquired by the learner from Junior School. It plays an important role in society through developing the learner's critical, creative and logical reasoning as well as analytical skills. Mathematics is critical in supporting other subjects for purposes of acquisition of competencies necessary for problem solving in day-to-day life. Mathematics is an essential element of communication. It enables one to explain, describe, illustrate, analyse, evaluate and represent information that helps in decision making processes. The subject is useful in computations particularly in Calculus, measurements, development of algebraic thinking and the study of geometry for the development of spatial relations as well as navigational skills. In addition, learners have the opportunity to study statistics and probability, which are critical in research and in understanding different population dynamics and phenomena. Learning activities that focus on the mathematical process skills of problem solving, communication, connection, collaboration, reasoning and proof as core to acquisition of the desirable competencies have been explicitly suggested. Key Pertinent and Contemporary Issues such as life skills, financial literacy, citizenship and education for sustainable development as well as values are integrated to improve the social and economic wellbeing of the individual and society.

Mathematics is a powerful tool in determining career choices in Science, Technology, Engineering and Mathematics (STEM), Social Sciences, Sports Science and Arts. These careers spread across medical fields, engineering, research, actuarial science, architecture, technology fields, computer and data science, and entrepreneurship, among others. The course is anchored on Sessional Paper no 1 of 2019 on policy framework for education and training for sustainable development, Vision 2030, which focuses on knowledge in science, technology and innovation and The Constitution of Kenya 2010 that stipulates that every child has a right to education. Further, in line with Africa Agenda 2063, the subject responds to the African Education Strategy, which focuses on strengthening the science and mathematics curricula.

GENERAL LEARNING OUTCOMES

By the end of Senior School the learner should be able to:

- a) Develop numerical computational skills, concepts and competencies for continuous learning and application of Mathematics,
- b) Collect, analyse and represent data for informed decision-making in real-life situations
- c) Evaluate and make financial decisions and transactions in real-life situations,
- d) Develop spatial skills and model geometric structures in real-life situations,
- e) Use digital technology for mathematical manipulations, explorations, innovations, navigational skills and for further learning,
- f) Develop critical and creative thinking, logical reasoning and analytical skills for problem solving, g) Use mathematical concepts and skills to support other learning areas for further training, research and career development, h) Develop confidence and interest in Mathematics as well as the coherence, utility and aesthetic nature of the subject for lifelong learning,
- i) Demonstrate appropriate responsiveness to pertinent and contemporary issues in the community and promote values for peaceful coexistence.

SUMMARY OF STRANDS AND SUB-STRANDS

	Strand	Sub-strand	Lessons
1	Numbers and Algebra	1.1 Real Numbers	8
		1.2 Indices and Logarithms	8
		1.3 Quadratic Expressions and Equations 1	16
2	Measurements and Geometry	2.1 Similarity and Enlargement	12
		2.2 Reflection and Congruence	12

		2.3 Rotation	10
		2.4 Trigonometry 1	15
		2.5 Area of Polygons	14
		2.6 Area of a Part of a Circle	12
		2.7 Surface Area and Volume of Solids	15
		2.8 Vectors 1	15
		2.9 Linear Motion	15
3	Statistics and Probability	3.1 Statistics I	16
		3.2 Probability 1	12
			180

Note: The suggested number of lessons per Sub-Strand may be less or more depending on the context.

STRAND 1.0: NUMBERS AND ALGEBRA**Sub-Strand 1: Real Numbers**

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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1.0 Numbers and Algebra	1.1. Real Numbers (8 lessons) ● <i>Classification of numbers</i> ● <i>Reciprocal of numbers</i>	By the end of the sub-strand, the learner should be able to: a) classify whole numbers as odd, even, prime and composite in different situations, b) classify real numbers as rational and irrational in different situations, c) determine the reciprocal of real numbers by division, d) determine the reciprocal of real numbers by use of mathematical tables and calculators, e) apply reciprocal of real numbers in mathematical computations,	The learner is guided to: ● discuss with peers to identify and categorise whole numbers as odd, even, prime and composite. ● discuss in a group the meaning of real numbers, ● categorise real numbers as rational and irrational numbers ● work out the reciprocal of real numbers by division, ● read the values of the reciprocals of real numbers from mathematical tables, ● work out the reciprocal of real numbers using a calculator ● use reciprocal of real numbers to work out tasks, ● use digital devices or other resources such as books to work	How do we use real numbers in day-to-day activities?
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		f) promote the use of real numbers in day-to-day activities.	out the reciprocal of real numbers.	
Core Competencies to be developed: • Communication and Collaboration: the learner discusses with peers to identify and categorise whole numbers as odd, even, prime, composite, rational, and irrational. • Critical Thinking and Problem Solving: the learner uses the reciprocal of real numbers to work out mathematical tasks. • Digital Literacy: the learner works out the reciprocal of real numbers from calculators and other digital devices.				
Values: • Unity: the learner discusses in a group and categorises real numbers as rational and irrational numbers. • Responsibility: the learner takes care of calculators and mathematical tables when working out the reciprocal of real numbers.				
Pertinent and Contemporary Issues (PCIs): Life Skills (self-esteem): the learner participates in the discussion on the concepts of rational and irrational numbers.				

Sub-Strand 2: Indices and Logarithms

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

1.0 Numbers and Algebra	1.2. Indices and Logarithms (8 lessons) • <i>Indices</i> • <i>Logarithms to base 10</i> • <i>Application of indices and logarithms</i>	By the end of the sub-strand the learner should be able to: a) express numbers in index form, b) derive the laws of indices using factors, c) apply the laws of indices in different situations, d) relate index notation to logarithm notation to base 10 e) determine common logarithms of numbers from mathematical tables and calculators, f) apply common logarithms in multiplication, division, powers and roots of numbers, g) appreciate the use of indices and common logarithms in mathematical computations.	The learner is guided to: • write numbers in terms of bases and indices and share them with peers. • discuss in a group and generate the laws of indices. • use the laws of indices to work out mathematical computations • discuss in a group and relate index notation to logarithm notation to base 10 • read and obtain common logarithms of numbers from tables and calculators. • use common logarithms of numbers to work out mathematical tasks. • use digital devices or other resources to work out	Why do we use indices and logarithms?
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			mathematical tasks involving common logarithms.	
Core Competencies to be developed: • Learning to learn: the learner discusses in a group and generates the laws of indices. • Digital literacy: the learner reads and obtains common logarithms of numbers from tables and calculators.				
Values: • Responsibility: the learner takes care of mathematical tables and calculators. • Respect: the learner appreciates the input of others while discussing in a group and relating powers of 10 to common logarithms.				
Pertinent and Contemporary Issues (PCIs): Life Skills – (self-awareness): the learner discusses in a group and relates powers of 10 to common logarithms and identifies the base.				

Sub-Strand 3: Quadratic Expressions and Equations

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

1.0 Numbers and Algebra	1.3 Quadratic Expressions and Equations (16 lessons) • <i>Quadratic expressions</i> • <i>Quadratic identities</i> • <i>Factorisation of quadratic expressions</i> • <i>Formation of quadratic equations</i> • <i>Solution of quadratic equations by factorisation</i> • <i>Application of quadratic equations to real-life situations</i>	By the end of the sub-strand, the learner should be able to: a) form quadratic expressions from different situations, b) derive the quadratic identities from the concept of area, c) apply quadratic identities in numerical cases, d) factorise quadratic expressions from different situations, e) form quadratic equations in different situations, f) solve quadratic equations by factorisation,	The learner is guided to: • generate quadratic expressions from given statements, • discuss with peers and come up with quadratic identities and write them on a chart. • discuss and use quadratic identities to evaluate numerical cases • discuss and work out factors of quadratic expressions. • work with peers to form quadratic equations from roots, statement and real life cases, • work out factors of quadratic equations to	How do we use equations in daily life?
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		g) apply quadratic equations to real life situations, h) explore the use of quadratic equations in real life situations.	obtain solutions to the equations, • search on the use of quadratic equations using digital devices or other resources.	
Core Competencies to be developed: • Critical thinking and problem solving: the learner interprets statements to form and solve quadratic equations to get solutions to real-life cases. • Communication and collaboration: the learner discusses with peers and comes up with quadratic identities.				
Values: • Unity: the learner works with others to write quadratic identities on a chart. • Respect: the learner works with peers to form and solve quadratic equations to get solutions to real-life cases.				
Pertinent and Contemporary Issues (PCIs): Socio-economic and environmental issues (social cohesion): the learner works with peers to form and solve quadratic equations to get solutions to real-life cases.				

Suggested Assessment Rubric: Numbers and Algebra

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
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Ability to classify whole numbers as odd, even, prime, and composite numbers, and real numbers as rational and irrational	The learner correctly and consistently classifies whole numbers as odd, even, prime, and composite numbers, and real numbers as rational and irrational	The learner correctly classifies whole numbers as odd, even, prime, and composite numbers, and real numbers as rational and irrational	The learner correctly classifies whole numbers as odd, even, prime, or composite numbers, or real numbers as rational and irrational	The learner correctly classifies whole numbers as odd, even, prime, or composite numbers,
Ability to determine the reciprocal of real numbers by division, tables and calculators	The learner consistently and accurately determines the reciprocal of real numbers by division, tables and calculators	The learner accurately determines the reciprocal of real numbers by division, tables and calculators	The learner accurately determines the reciprocal of real numbers by division or tables or calculators	The learner partially determines the reciprocal of real numbers by division, or tables or calculators
Ability to apply reciprocal of real numbers in mathematical computations	The learner systematically and accurately applies the reciprocal of real numbers in mathematical computations	The learner accurately applies the reciprocal of real numbers in mathematical computations	The learner partially applies the reciprocal of real numbers in mathematical computations	The learner partially applies the reciprocal of real numbers in mathematical computations with prompts

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to derive and apply laws of indices.	The learner systematically and accurately derives and applies laws of indices	The learner accurately derives and applies laws of indices	The learner accurately derives or applies laws of indices	The learner has challenges deriving and applying laws of indices
Ability to determine common logarithms of numbers from mathematical tables and calculators	The learner consistently and accurately determines common logarithms of numbers from mathematical tables and calculators	The learner accurately determines common logarithms of numbers from tables and calculators	The learner accurately determines common logarithms of numbers from mathematical tables or calculators	The learner has challenges determining common logarithms of numbers from mathematical tables and calculators
Ability to apply common logarithms in multiplication, division, powers, and roots of numbers	The learner appropriately and accurately applies common logarithms in multiplication, division, powers, and roots of	The learner accurately applies common logarithms in multiplication, division, powers, and roots of	The learner accurately applies common logarithms in multiplication or division or powers or roots of	The learner has challenges applying common logarithms in multiplication, division, powers and roots of numbers

	numbers	numbers	numbers	
Ability to form quadratic expressions	The learner consistently and correctly forms quadratic expressions	The learner correctly forms quadratic expressions	The learner partially correctly forms quadratic expressions	The learner has challenges forming quadratic expressions

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Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to derive quadratic identities from the concept of area	The learner consistently and correctly derives quadratic identities from the concept of area	The learner correctly derives quadratic identities from the concept of area	The learner partially correctly derives quadratic identities from the concept of area	The learner has challenges deriving quadratic identities from the concept of area
Ability to factorise quadratic expressions	The learner systematically and correctly factorises quadratic expressions	The learner correctly factorises quadratic expressions	The learner partially correctly factorises quadratic expressions	The learner has challenges factorising quadratic expressions

Ability to form, solve, and apply quadratic equations in real-life situations	The learner systematically and accurately forms, solves, and applies quadratic equations in real-life situations	The learner accurately forms, solves, and applies quadratic equations in real-life situations	The learner partially accurately forms solves, and applies quadratic equations in real-life situations	The learner has challenges forming, solving, and applying quadratic equations in real-life situations
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STRAND 2.0: MEASUREMENTS AND GEOMETRY

Sub-Strand 1: Similarity and enlargement

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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2.0 Measurements and Geometry	2.1 Similarity and Enlargement (12 lessons) • <i>Similarity</i> • <i>Enlargement</i> • <i>Scale Factors</i> • <i>Application of Similarity and Enlargement</i>	By the end of the sub strand the learner should be able to: a) determine the centre of enlargement and the linear scale factor for similar figures, b) construct the image of an object under an enlargement given the centre and the linear scale factor, c) determine the area and volume scale factor of different figures and solids, d) relate linear scale factor, area scale factor, and volume	The learner is guided to: • discuss in a group and review the properties of similar figures and enlargement, • use an object and its image to establish the centre of enlargement and the ratio of the lengths of corresponding sides, Linear Scale Factor (L.S.F) • discuss in a group and draw on a plane surface the images of objects under enlargement given the centres and linear scale factors • draw on the Cartesian plane the images of	How are similarity and enlargement applied in day-to day life?
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		scale factor in enlargements e) apply similarity and enlargement to real life situations, f) appreciate the use of similarity and enlargement in real life situations.	objects under enlargement given the centres and linear scale factors. • discuss in a group and establish Linear Scale Factor (L.S.F) from similar plane figures. Work out the ratio of the area of similar plane figures, Area Scale Factor (A.S.F). Work out the ratio of volume of similar solids, Volume Scale Factor (V.S.F) • discuss in a group and establish the relationship between L.S.F, A.S.F and V.S.F using two similar	
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			solids. • work out tasks involving similarity and enlargements in real-life situations. • use digital devices and other resources to learn	
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			more on the use and application of similarity and enlargement Project Use and improvise locally available materials to make sets of models of solids such as cubes, cuboids, cylinders and spheres of different sizes using similarity and enlargement to be used as learning resources in	
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			school.	
Core Competencies to be developed: • Communication and Collaboration: the learner discusses and recognises similar objects from their immediate environment. • Creativity and Imagination: the learner makes models of solids of different sizes using similarity and enlargement concepts.				
Values: • Responsibility: the learner collects and takes care of objects from the immediate environment to recognise those that are similar and discusses what makes them similar. • Integrity: the learner makes models with accurate measurements, adhering to the concept of similarity and enlargement.				
Pertinent and Contemporary Issues (PCIs): • Social Cohesion: the learner works in a group to create models of various similar solids. • Education for Sustainable Development (environmental education): the learner collects and uses objects from the immediate environment and discusses with peers to recognise those that are similar.				

Sub-Strand 2: Reflection and congruence

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

2.0 Measurements and Geometry	2.2 Reflection and Congruence (12 Lessons) • <i>Symmetry</i> • <i>Reflection</i> • <i>Equations of Mirror Lines</i> • <i>Congruence</i>	By the end of the sub strand, the learner should be able to: - identify lines of symmetry in plane figures, - determine the properties of reflection in different situations - draw an image given an object and a mirror line on plane surface and Cartesian plane, - determine the equation of the mirror line given an object and its image, - carry out	The learner is guided to: - collect and observe different objects from the immediate environment and illustrate the lines and planes of symmetry, - use a plane mirror to locate the image of an object placed at a point in front of the mirror, - draw a plane figure and a mirror line on tracing paper. Fold the paper along the mirror line and trace out the plane figure. Use the two plane figures and the mirror line to generate the properties of reflection, - draw on plain paper a given object and a mirror line. Use the properties of reflection to locate the corresponding image,	- How do we use reflection in day-to-day life? - Where do we use congruence in real life?
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		congruence tests for triangles, f) promote use of reflection and congruence in real life situations.	• draw on a Cartesian plane given objects and mirror lines. Use the properties of reflection to draw the corresponding images, • construct a mirror line given an object and its image on a Cartesian plane. Work out the equation of the mirror line, • work in a group and make paper cutouts of different identical shapes. Use the shapes to identify direct and opposite congruent shapes, • use different triangles to establish the congruence tests; three sides (SSS), two sides and an included angle (SAS), two angles and a side (AAS or SAA or ASA), and right angle, hypotenuse and side (RHS) • discuss the applications of reflections and congruence on images formed by driving mirrors to enhance road safety,	
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			• use digital devices to carry out reflection of different objects to generate images.	
Core Competencies to be developed: • Learning to Learn: the learner applies the properties of reflection to generate the image of an object on a given mirror line. • Creativity and Imagination: the learner locates the image of an object placed at a point in front of a mirror.				
Values: • Responsibility: the learner takes care of the mirror used in reflecting corresponding images. • Respect: the learner recognises the input of every member of the team as they work in groups to make paper cutouts of different identical shapes to identify direct or opposite congruent shapes.				
Pertinent and Contemporary Issues (PCIs): • Road safety: the learner discusses with peers the applications of reflections and congruence on images formed by driving mirrors in enhancing road safety. • Environment and Technology: the learner selectively and cautiously collects and observes different objects from the immediate environment and identifies the lines and planes of symmetry.				

Sub-Strand 3: Rotation

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurements and Geometry	2.3 Rotation (10 lessons) • <i>Properties of rotation</i> • <i>Rotation on different planes</i> • <i>Rotational symmetry</i> • <i>Rotation and Congruence</i>	By the end of the sub strand the learner should be able to: a) determine the properties of rotation in different situations, b) rotate an object given the centre and angle of rotation on a plane surface and the Cartesian plane, c) determine the centre and angle of rotation given an object and its image, d) determine the	The learner is guided to: • demonstrate rotation using an actual or improvised clock. Discuss and explain the movement of the hour or minute hand • use an object and its image on a plane surface to discuss and generate the properties of rotation. • generate an image of an object given a centre and angle of rotation. • carry out rotation on a Cartesian plane given the object, centre, and angle of rotation • in a group, use construction to find the centre and angle of rotation given the object and its image on a plane surface and the Cartesian plane,	How is rotation applied in real-life situations?

		order of rotational symmetry of plane figures,	• use paper cutouts of different plane figures to locate points of	
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		e) determine the axis and order of rotational symmetry in solids, f) deduce congruence from rotation, g) appreciate the application of rotation in real-life situations.	symmetry and establish the order of rotational symmetry, • collect regular solids such as pyramids, triangular prisms, cones, tetrahedrons from the immediate environment and identify the axis to establish the order of rotational symmetry, • use different objects and their images to identify the type of congruence in rotation, • use digital devices and other resources to learn more on rotation of plane figures and solids.	
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Core competencies to be developed:

- Critical thinking and Problem Solving: the learner draws objects and their images and through construction establishes the centres and angles of rotation.
- Communication and Collaboration: the learner discusses using an object and its image on a plane surface to generate the properties of rotation.

Values:

- Responsibility: the learner takes care of learning resources collected from the immediate environment to establish their axis and order of rotational symmetry.
- Respect: learners appreciate each other's opinion as they discuss and draw the image under a rotation given the object, centre and angle of rotation.

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Pertinent and Contemporary Issues (PCIs):

Environmental education: as the learner responsibly collects regular solids from the immediate environment to determine the axis and order of rotational symmetry.

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Sub-Strand 4: Trigonometry 1

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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2.0 Measurements and Geometry	2.4 Trigonometry 1 (15 lessons) • <i>Trigonometric ratios of acute angles</i> • <i>Sines and Cosines of complementary angles</i> • <i>Trigonometric ratios of special angle</i> • <i>Application of trigonometric ratios</i>	By the end of the sub strand, the learner should be able to: - determine the trigonometric ratios of acute angles from mathematical tables and calculators, - relate sines and cosines of complementary angles, - relate the sine, cosine and tangent of acute angles, - determine trigonometric ratios of special angles: 30°, 45°, 60° and 90° using triangles,	The learner is guided to: - identify from the immediate environment shapes that make right-angled triangles. • draw right-angled triangles and use them to define and work out sine, cosine, and tangent ratios, - use mathematical tables and calculators to obtain trigonometric ratios, - generate a table of angles and their complements and determine their sines and cosines. Make observations and establish the relationships between the sine and cosine ratios, - work in a group and using different acute angles, generate a table of the ratios	- What is trigonometry? - How do we use trigonometry in real-life situations?
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		e) apply trigonometric ratios to angles of elevation and depression, f) reflect on the use of trigonometry in real life situations.	of sine, cosine, and tangents to establish the relationship between sine, cosine, and tangent, • work in a group and use an isosceles right-angled triangle and an equilateral triangle to generate the trigonometric ratios of special angles 30°, 45°, 60° and 90°, • use trigonometric ratios to work out tasks related to triangles, • use a simple clinometer to measure the angle of elevation and depression from a target object in the immediate environment, • use trigonometric ratios to work out tasks involving	
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			angles of elevation and depression, • use digital devices and other resources such as books,	
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			manuals, and journals to learn more about trigonometric ratios involving triangles, angles of depression, and elevation. Project Make a simple clinometer using locally available materials and use it as a learning resource	
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Core Competencies to be developed:

- Creativity and Imagination: the learner makes a simple clinometer using locally available material and uses it as a learning resource
- Critical Thinking and Problem Solving: the learner discusses and makes observations about the sine and cosine of complementary angles, and applies trigonometric ratios in the solution of triangles
- Digital literacy: the learner uses digital devices and other resources such as books, manuals, and journals to learn more about trigonometric ratios

Values:

- Responsibility: the learner takes care of mathematical tables and calculators used in determining trigonometric ratios
- Unity: the learner works in a group, for instance, when using an isosceles right-angled triangle and an equilateral triangle to generate the trigonometric ratios of special angles 30° , 45° , 60° , and 90°

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Pertinent and Contemporary Issues (PCIs):

- Environmental Education and Protection: the learner identifies shapes in the immediate environment that have right-angled triangles, and makes a simple clinometer using materials from the immediate environment
- Parental Engagement and Empowerment: the learner gets support from parents/guardians to get resources to make a clinometer.

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Sub-Strand 5: Area of Polygons

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurements and Geometry	2.5 Area of Polygons (14 lessons) - Area of a triangle using $\frac{1}{2} \times \text{base} \times \text{height}$ - Area of a triangle using Heron's formula - Area of quadrilaterals - Area of regular and irregular polygons	By the end of the sub-strand the learner should be able to: - derive the formula for the area of a triangle given two sides and an included angle, - work out the area of a triangle given two sides and an included angle, - determine the area of a triangle using Heron's formula, - determine the area of quadrilaterals in different situations, - work out the area of regular heptagon and octagon, - determine the area of irregular	The learner is guided to: - discuss in groups and use trigonometric ratios to generate the formula for the area of a triangle given two sides and an included angle. For example, $\frac{1}{2} \times \text{side}_1 \times \text{side}_2 \times \sin(\text{included angle})$ - use the formula $\frac{1}{2} \times \text{base} \times \text{height}$ to calculate the area of different triangular shapes, - use the Heron's Formula (Heron's Formula) to work out the areas of various triangles, - identify different quadrilaterals and work out their areas, - identify from the immediate environment objects with the	- How do we work out the area of polygons? - How do we apply the concept of the area of polygons in real-life situations?

		polygons in different situations,	shape of regular polygons and calculate the area of regular polygons up to heptagon,	
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	• <i>Application of area of polygons</i>	g) explore the area of polygons as used in real life situations.	• use objects from the immediate environment that have shapes of irregular polygons, discuss and work out their area, • research and discuss in a group the use of the area of polygons in real-life situations.	
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Core Competencies to be developed:

- Critical Thinking and Problem Solving: the learner uses trigonometric ratios to generate the formula for a triangle given two sides and an included angle.
- Creativity and Imagination: the learner identifies from the immediate environment objects in the shape of regular polygons and calculates their area.
- Learning to Learn: the learners research and discuss the use of the area of polygons in real-life situations.

Values:

- Patriotism: the learner uses the objects from the environment and takes care of the environment.
- Responsibility: the learner identifies objects from the immediate environment to use in calculating the area of regular polygons.
- Peace: the learner harmoniously discusses in groups and uses trigonometric ratios to generate the formula for the area of a triangle given two sides and an included angle.

Pertinent and Contemporary Issues (PCIs):

Environmental Education: the learner collects objects from the immediate environment to be used as learning materials for area of polygons.

Sub-Strand 6: Area of a Part of a Circle

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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2.0 Measurements and Geometry	2.6 Area of a Part of a Circle (12 lessons) • <i>Area of an annulus</i> • <i>Area of a sector of a circle</i> • <i>Area of an annular sector</i> • <i>Area of a segment</i> • <i>Area of common region between two intersecting circles</i>	By the end of the sub strand the learner should be able to: a) determine the area of an annulus in different situations, b) work out the area of a sector of a circle in real-life situations, c) determine the area of an annular sector in different situations, d) work out the area of a segment of a circle, e) determine the area of common region between two intersecting circles, f) apply the area of a part of a circle in real-life situations,	The learner is guided to: • use circular shapes or objects to identify concentric rings formed by inner and outer space. Hence work out the area of an annulus, • work in a group, and use paper cutouts to make sectors of circles to determine their areas as part of a circle, • use different examples to illustrate and work out the area of an annular sector, for example area covered by a car wiper blade when cleaning the windscreen, • draw different segments of a circle and calculate their area, • draw illustrations of two intersecting circles to identify the common area/part between the two circles. Work out the area of common area/part.	How do we use the concept of the area of a part of a circle in real life?
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	● <i>Application on the use of the area of a part of a circle</i>	g) explore the use of the area of a part of a circle in real-life situations.	• relate and work out the area of a part of a circle in real-life situations, • use digital devices and other resources to learn more about the area of a part of a circle. Project • make a dartboard of different numbers of concentric circles from locally available materials. Paint different colours on the annular sectors and the concentric circles. Discuss and create rules for playing and scoring the game. • make necklaces using beading techniques of different coloured annular sectors.	
Core Competencies to be developed: • Creativity and Imagination: the learner makes an improvised dartboard of different numbers of concentric circles. • Learning to Learn: the learner draws illustrations of two intersecting circles to identify the common area/part between the two circles. • Communication and collaboration: the learners engage to improvise locally available materials to make dartboards and beaded necklaces.				

Values:

- Responsibility: the learner makes dartboard of different numbers of concentric circles from the immediate environment
- Unity: the learner collaborates in a group and uses paper cutouts to make sectors of circles to determine their areas as part of a circle

Pertinent and Contemporary Issues (PCIs):

- Parental Engagement and Empowerment: the learner engages parents in using locally available materials for making improvised beaded necklaces.
- Environmental Education: the learner sources locally available materials to make necklaces and dartboards.
- Learner Support Programmes (Peer Education): the learner works with others to make necklaces using beading techniques of different coloured annular sectors.

Sub-Strand 7: Surface Area and Volume of Solids

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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2.0 Measurements and Geometry	2.7 Surface Area and Volume of Solids (15 lessons) • <i>Surface Area of Solids</i> • <i>Volume of Solids</i>	By the end of the sub strand, the learner should be able to: a) determine the surface area of prisms, pyramids, cones, frustums and spheres, b) calculate the surface area of composite solids, c) calculate the volume of prisms, pyramids, cones, frustums and spheres, d) determine the volume of composite solids, e) explore the use of the surface area and volume of solids in real-life situations.	The learner is guided to: • use models and nets of prisms, pyramids, cones, frustums, and spheres to calculate surface areas. • identify the components of a composite solid, calculate the surface area of each component and obtain the total surface area of the composite solid. • use models of a cone and a cylinder of the same height and base area and an appropriate substance such as sand or water, to demonstrate that the volume of the cone is a third of the volume of the cylinder.	1. How do we determine the surface area and volume of solids? 2. Why do we determine the surface area and volume of solids?
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			• discuss in a group and work out the volume of prisms, pyramids, cones, frustums, and spheres. • discuss in a group the components of composite solids, work out the volume of the components and obtain the total volume. • use appropriate containers from the local environment to work out the volume and capacity of the prism, pyramid, cone, frustum and sphere. • use digital devices and other resources to work out the surface area and volume of solids.	
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Core Competencies to be developed:

- Critical Thinking and Problem Solving: the learner uses a cylinder and a cone of the same radius and height, and a substance such as sand or water, to determine the volume of a cone as a third of the volume of a cylinder.
- Creativity and Imagination: the learner uses a cylinder and a cone of the same radius and height, and a substance such as sand or water, to determine the volume of a cone as a third the volume of a cylinder.

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Values:

- Unity: the learner discusses in a group and establishes the relationship between the volume of a cone and the volume of a cylinder.
- Respect: the learner listens to others' opinions as they discuss in groups and work out the volumes of prisms, pyramids, cones, frustums and spheres.

Pertinent and Contemporary Issues (PCIs):

Environment and technology: as the learner collects and uses containers from the immediate environment and uses them to discuss and work out their volumes.

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Sub-Strand 8: Vectors I

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
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2.0 Measurements and Geometry	2.8 Vectors I (15 lessons) • <i>Vector and scalar quantities.</i> • <i>Vector notation</i> • <i>Representation of vectors</i> • <i>Equivalent vectors</i> • <i>Addition of vectors</i> • <i>Multiplication of vectors by scalar</i> • <i>Column vectors</i> • <i>Position vectors</i> • <i>Magnitude of a vector</i> • <i>Midpoint of a vector</i>	By the end of the sub strand the learner should be able to: a) explain vector and scalar quantities with illustrations b) use vector notations in different ways c) represent vectors geometrically in different situations d) identify equivalent vectors in different situations e) add vectors in different situations f) multiply vectors by scalars g) determine column vectors	The learner is guided to: • search and brainstorm on the meaning of the terms vector and scalar quantities and illustrate giving examples, • practise writing vector notations on charts • illustrate vectors on charts, grids and regular plane figures, • discuss the meaning of arrows and lines on roads and road signs in relation to movement on the roads to enhance safety, • demonstrate equivalent vectors on charts, grids and Cartesian coordinates,	1. What are vectors? 2. Where are vectors used in real-life situations?
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		in different situations		
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	• <i>Translation vector</i>	h) determine position vectors in different situations i) work out magnitude of a vector in different situations j) determine the midpoint of a vector in different situations k) determine translation vector as a transformation l) appreciate the use of vectors in real-life situations	• illustrate addition of vectors using charts and grids to get the resultant vector, • illustrate multiplication of vectors by a scalar on charts and grids, • express vectors in column form on Cartesian planes, grids and charts, • discuss and illustrate position vectors using any arbitrary reference points like the origin, • work out magnitude of different vectors, • find the midpoint of different vectors, • use vectors to illustrate translation as a transformation, • use digital devices to watch videos and	
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			play games involving vectors.	
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Core competencies to be developed:

- Learning to learn: As the learner uses charts and grids to practise addition of vectors.
- Self-efficacy: As the learner shares and uses resources from the immediate environment such as plane figures or flip charts to demonstrate equivalent vectors.

Values:

- Responsibility: As the learner uses resources from the immediate environment such as plane figures, flip charts and relate equivalent vectors.
- Unity: As the learner searches and brainstorms on the meaning of the terms vector and scalar quantities and illustrates giving examples.

Pertinent and Contemporary Issues (PCIs):

- Education for Sustainable Development - Environmental education as learners practice addition of vectors using charts and grids made from resources in the immediate environment.
- Road safety: as the learner discusses the meaning of arrows and lines on roads and road signs in relation to movement on the roads to enhance safety.

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Sub-Strand 9: Linear Motion

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurements and Geometry	2.9 Linear Motion (15 lessons) - Distance, displacement, speed, velocity, and acceleration - Velocity and acceleration - Displacement time graph - Interpretation of displacement time graph - Velocity time graph	By the end of the sub strand, the learner should be able to: a) explain the terms distance, displacement, speed, velocity and acceleration in real life situations, b) determine velocity and acceleration in different situations, c) draw a displacement time graph of different situations, d) interpret displacement-time graph in different	The learner is guided to: - demonstrate motion by moving an object from a point to another for a specified distance and direction and state the meaning of the terms distance, displacement, speed, velocity and acceleration, - participate in a race of a specified distance. Record time taken by each learner and work out the velocity and acceleration of different learners, - work out the velocity and acceleration of movement of	- What is speed? - How do we use displacement time graphs in real life? - How do we use velocity time graphs?

		situations, e) draw a velocity-time graph from given	objects from one point to another,	
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	• <i>Interpretation of velocity - time graph</i> • <i>Relative speed</i>	tables in different situations, f) interpret velocity time graph in different situations, g) determine relative speed of two moving bodies in different situations, h) appreciate use of linear motion in real life situations.	• use data of displacement and time to plot displacement time graphs on a grid or Cartesian plane, • obtain information from a graph and examine it in relation to displacement, time and velocity, • use data of velocity and time taken to draw a velocity-time graph on a grid or Cartesian plane, • obtain information from a graph and examine it in relation to velocity, time and acceleration, • observe two vehicles when one is overtaking, discuss and work out the relative speed of the two vehicles. Also discuss the dangers of careless overtaking • use digital devices and other resources to explore more on linear motion in	
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			day-to-day life.	
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Core Competencies to be developed:

- Communication and Collaboration: the learner discusses different displacement-time graphs.
- Self-Efficacy: the learner participates in a race of a specified distance, record time taken by a learner and work out the velocity and acceleration.
- Critical Thinking and Problem Solving: the learner observes two bodies in motion such as cars when overtaking or moving in opposite directions and works out the relative speed of the two bodies in motion.

Values:

- Responsibility: as the learner uses objects in motion such as thrown objects and many others to explain distance, displacement, speed, velocity and acceleration.
- Integrity: as the learner participates in a race of a specified distance, records time taken by each learner and accurately works out the velocity and acceleration of different learners.
- Unity: as the learner participates and collaborates in a race to record time taken by each learner and work out the velocity and acceleration of different learners.

Pertinent and contemporary Issues (PCIs):

- Learner Support Programmes (Sports and Games): the learner participates in a race of a specified distance, records time taken by each learner and works out the velocity and acceleration,
- Socio-Economic and Environmental Issues (Safety and Security): the learner observes the two bodies in motion such as cars when overtaking or moving in opposite directions and works out the relative speed of these two bodies in motion. • Road safety: as the learner observes two vehicles when one is overtaking, discusses and works out the relative speed of the

two vehicles. Also discusses the dangers of careless overtaking.

Suggested Assessment Rubric: Measurements and Geometry

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to determine the centre of enlargement and the linear scale factor from similar figures	The learner systematically and accurately determines the centre of enlargement and linear scale factor from similar figures	The learner accurately determines the centre of enlargement and linear scale factor from similar figures	The learner needs assistance to accurately determine the centre of enlargement and linear scale factor from similar figures	The learner partially determines the centre of enlargement or linear scale factor from similar figures
Ability to construct the image of an object under an enlargement given the centre and the linear scale factor	The learner accurately and precisely constructs objects and their images under enlargement given the centre and the	The learner accurately constructs objects and their images under enlargement given the centre	The learner partially accurately constructs objects and their images under enlargement given the centre and the linear scale factor	The learner partially constructs objects or their images under enlargement given the centre and the linear scale factor

	linear scale factor	and the linear scale factor		
Ability to carry out enlargement on the Cartesian plane	The learner accurately and systematically carries out enlargement on the Cartesian plane	The learner accurately carries out enlargement on the Cartesian plane	The learner partially accurately carries out enlargement on the Cartesian plane partly accurately	The learner partially carries out enlargement on the Cartesian plane
Ability to determine linear, area and	The learner precisely and correctly determines linear, area	The learner correctly determines linear, area and	The learner correctly determines either linear	The learner determines linear or area scale factors

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
volume scale factors	and volume scale factors	volume scale factors	or area or volume scale factors	

Ability to apply similarity and enlargement to real life situations	The learner systematically and correctly applies similarity and enlargement to real-life situations	The learner correctly applies similarity and enlargement to real-life situations	The learner needs assistance to correctly partially correctly applies similarity and enlargement to some real life situations	The learner partially applies similarity and enlargement to real life situations
Ability to identify lines of symmetry of plane figures and properties of reflection	The learner precisely and correctly identifies lines of symmetry of plane figures and properties of reflection	The learner correctly identifies lines of symmetry of plane figures and properties of reflection	The learner correctly identifies lines of symmetry of plane figures or properties of reflection	The learner partially identifies lines of symmetry of plane figures or properties of reflection
Ability to draw an image given an object and a mirror line on a plane surface and Cartesian plane and determine	The learner systematically and accurately draws an image given an object and a mirror line on a plane surface and Cartesian plane and determines	The learner accurately draws an image given an object and a mirror line on a plane surface and Cartesian plane	The learner accurately draws an image given an object and a mirror line on a plane surface or Cartesian plane or determines the equation of the mirror line	The learner partially draws an image given an object and a mirror line on the plane surface or the Cartesian plane or determines the equation of the mirror

the equation of the mirror line	the equation of the mirror line	and determines the equation of the mirror line		line
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Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to determine the properties of rotation and rotate an object, given the centre and angle of rotation	The learner comprehensively and correctly determines the properties of rotation and rotates an object, given the centre and angle of rotation	The learner correctly determines the properties of rotation and rotates an object, given the centre and angle of rotation	The learner correctly determines the properties of rotation or rotates an object, given the centre and angle of rotation	The learner partially determines the properties of rotation or rotates an object, given the centre and angle of rotation
Ability to determine the order of rotational symmetry in solids and relate	The learner comprehensively and correctly determines the order of rotational symmetry in solids and relates	The learner correctly determines the order of rotational symmetry in	The learner correctly determines the order of rotational symmetry in solids or relates rotation and congruence	The learner partially determines the order of rotational symmetry in solids or relates rotation and congruence

rotation and congruence	rotation and congruence	solids and relates rotation and congruence		
Ability to determine the trigonometric ratios of acute angles from right angled triangles, mathematical tables, and calculators	The learner systematically and accurately determines the trigonometric ratios of acute angles from right-angled triangles, mathematical tables, and calculators	The learner accurately determines the trigonometric ratios of acute angles from right-angled triangles, mathematical tables, and calculators	The learner partially determines the trigonometric ratios of acute angles from right angled triangles, mathematical tables, and calculators	The learner determines the trigonometric ratios of acute angles from right-angled triangles, or mathematical tables, or calculators with assistance

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
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Ability to relate the sines, cosines, of complementary angles and sines, cosines, and tangents of acute angles	The learner logically and accurately relates the sines, cosines, of complementary angles and sines, cosines, and tangents of acute angles	The learner accurately relates the sines, cosines, of complementary angles and sines, cosines, and tangents of acute angles	The learner accurately relates the sines, cosines, of complementary angles or sines, cosines, or tangents of acute angles	The learner partially relates the sines, cosines, of complementary angles or sines, cosines, and tangents of acute angles
Ability to determine trigonometric ratios of special angles 30^0 , 45^0 , 60^0 , and 90^0 using triangles,	The learner systematically and accurately determines trigonometric ratios of special angles 30^0 , 45^0 , 60^0 , and 90^0 using triangles	The learner accurately determines trigonometric ratios of special angles 30^0 , 45^0 , 60^0 , and 90^0 using triangles	The learner accurately determines trigonometric ratios of any two of the special angles 30^0 , 45^0 , 60^0 , and 90^0 using triangles,	The learner partially determines trigonometric ratios of any one of the special angles 30^0 , 45^0 , 60^0 , and 90^0 using triangles

Ability to apply trigonometric ratios in the solution of triangles, angles of elevation, and depression	The learner appropriately and accurately applies trigonometric ratios in the solution of triangles, angles of elevation, and depression	The learner accurately applies trigonometric ratios in the solution of triangles, angles of elevation, and depression	The learner accurately applies trigonometric ratios in any two of either the solution of triangles or angles of elevation or depression	The learner partially applies trigonometric ratios in any one of either solution of triangles or angles of elevation or depression
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Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to determine the area of quadrilaterals, heptagons, octagons, and irregular polygons	The learner systematically and accurately determines the area of quadrilaterals, heptagons, octagons, and irregular polygons	The learner accurately determines the area of quadrilaterals, heptagons, octagons, and irregular polygons	The learner accurately determines the area of quadrilaterals, heptagons, octagons, or irregular polygons	The learner partially determines the area of quadrilaterals, heptagons, octagons or irregular polygons

Ability to determine the area of an annulus, sector, annular sector, and segment of a circle	The learner systematically and accurately determines the area of an annulus, sector, annular sector, and segment of a circle	The learner accurately determines the area of an annulus, sector, annular sector, and segment of a circle	The learner accurately determines the area of an annulus, or sector, annular sector, or segment of a circle	The learner partially determines the area of an annulus or sector or annular sector or segment of a circle
Ability to determine the area of a common region between two intersecting circles	The learner systematically and correctly determines the area of the common region between two intersecting circles	The learner correctly determines the area of the common region between two intersecting circles	The learner partially correctly determines the area of the common region between two intersecting circles	The learner partially determines the area of the common region between two intersecting circles
Ability to determine the surface area of prisms, pyramids,	The learner systematically and accurately determines	The learner accurately determines the	The learner accurately determines the surface area of prisms,	The learner partially determines the surface area of prisms,

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
cones, frustums and spheres, and composite solids	the surface area of prisms, pyramids, cones, frustums, spheres, and composite solids	surface area of prisms, pyramids, cones, frustums, spheres, and composite solids	pyramids, cones, frustums, spheres, or composite solids	pyramids, cones, frustums, spheres, or composite solids.
Ability to determine the volume of prisms, pyramids, cones, frustums, spheres, and composite solids	The learner consistently and correctly determines the volume of prisms, pyramids, cones, frustums, spheres, and composite solids	The learner correctly determines the volume of prisms, pyramids, cones, frustums, spheres, and composite solids	The learner correctly determines the volume of prisms, pyramids, cones, frustums, spheres, or composite solids	The learner partially determines the volume of prisms, pyramids, cones, frustums, spheres, or composite solids

Ability to represent vectors geometrically, perform operations on vectors and determine the magnitude of vectors	The learner precisely and correctly represents vectors geometrically, performs operations on vectors and determines the magnitude of vectors	The learner correctly represents vectors geometrically, performs operations on vectors and determines the magnitude of vectors	The learner correctly represents vectors geometrically, performs operations on vectors or determines the magnitude of vectors	The learner partially represents vectors geometrically, performs operations on vectors or determines the magnitude of vectors
Ability to determine translation vector	The learner systematically and accurately determines translation vector	The learner accurately determines translation vector	The learner partially accurately determines translation vector	The learner partially determines translation vector

Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
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Ability to determine displacement, speed, velocity, acceleration;	The learner explicitly and correctly determines displacement, speed, velocity, and acceleration	The learner correctly determines displacement, speed, velocity, and acceleration	The learner correctly determines displacement or speed or velocity or acceleration	The learner partially determines displacement or speed or velocity or acceleration
Ability to draw and interpret graphs of linear motion	The learner systematically and accurately draws and interprets graphs of linear motion	The learner accurately draws and interprets graphs of linear motion	The learner accurately draws or interprets graphs of linear motion	The learner partially draws or interprets graphs of linear motion
Ability to solve problems involving relative speed	The learner correctly and systematically solves problems involving relative speed	The learner correctly solves problems involving relative speed	The learner partially correctly solves problems involving relative speed	The learner partially solves problems involving relative speed

Sub Strand 1: Statistics and Probability

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Statistics and Probability	3.1. Statistics I (16 lessons lessons) • <i>Collection of Data</i> • <i>Frequency distribution table</i> • <i>Mean, Mode and Median of grouped and ungrouped data</i> • <i>Representation of Data</i> • <i>Interpretation of Data</i>	By the end of the sub strand the learner should be able to: a) collect data from real life sources, b) draw a frequency distribution table for grouped and ungrouped data, c) determine mean, mode and median of grouped and ungrouped data, d) represent data using histograms and frequency polygons, e) interpret data from histograms and frequency polygons, f) promote data	The learner is guided to: • discuss with peers the meaning of the term data. Explore data sources and use different and appropriate methods to collect and record data, • collect data on availability of pedestrian crossing points along major roads in the locality or collect any other data and use it to prepare a frequency distribution table for an ungrouped data, • group collected data or any other data into appropriate classes and generate a frequency distribution table, • discuss in pairs and identify the modal class for grouped data, • use data collected from different sources and work out the mode,	1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life?

		collection,		
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		organisation and representation for informed decision making.	median and mean for grouped and ungrouped data, • work in a group and use collected data to draw histograms (<i>with equal and unequal class-width</i>) and frequency polygons, • use the histograms and frequency polygons to examine data to make conclusions, Project Identify a situation or problem in the school or community. Collect data on it, organise the data and represent it using any methods to guide the school or community to make informed decisions.	
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Core Competencies to be developed:

- Critical thinking and problem solving: the learner uses the histograms and frequency polygons to examine data and make conclusions.
- Self-Efficacy: the learner discusses with peers the meaning of the term data and explores data sources.
- Digital literacy: the learner uses digital devices to draw histograms and frequency polygons to make conclusions.
- Communication and collaboration: learner presents and shares findings from data to different groups of people such as school administrators, parents or local community leaders.

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Values:

- Integrity: the learner collects and records data accurately.
- Unity: the learner works in groups and uses collected data to draw histograms and frequency polygons.
- Love: the learner is considerate of the other as they discuss in pairs and identify the modal class for grouped data

Pertinent and Contemporary Issues (PCIs):

- Life Skills (Decision Making): the learner makes informed decisions from data interpretation from histograms and frequency polygons.
- Parental Engagement and Empowerment: the learner seeks support and guidance from parents or guardians to collect, organise and present data on an identified situation in the school or community.

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Sub-Strand: Probability 1

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Statistics and Probability	3.2 Probability 1 (12 lessons) • <i>Experimental probability</i> • <i>Probability space</i> • <i>Mutually exclusive and independent events</i> • <i>Laws of Probability</i> • <i>Probability tree diagrams</i>	By the end of the sub strand the learner should be able to: a) perform experiments involving probabilities in different situations, b) identify the range of probability in different situations, c) generate probability space of different events, d) determine the probability of mutually exclusive and independent events, e) apply the laws of probability in different situations,	The learner is guided to: • carry out experiments such as tossing coins or throwing dice to illustrate experimental probability, • discuss with peers the probabilities of events and show the range of probability, • list down all the possible outcomes of different events in real-life situations such as getting pedestrians crossing the roads at undesignated areas and highlight its dangers for public awareness, • work out the probability of mutually exclusive events and independent events, • work out probability of events using addition and multiplication law • in pairs, use tree diagram to work out probability of two independent and mutually	How is probability applied in real life situations?

		f) determine the probability of	events,	
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		independent events using tree diagrams, g) appreciate the application of probability in real-life situations.	• discuss, create and play games related to probability, • use digital devices and other resources to learn more on probability.	
Core competencies to be developed: • Communication and collaboration: the learner carries out experiments such as tossing coins or throwing dice to illustrate experimental probability. • Citizenship: the learner acknowledges Kenyan currency as they carry out experiments of tossing coins. • Self-efficacy: the learner appreciates his/her ability to play games related to probability.				
Values: • Responsibility: the learner carefully tosses coins or throws dice to illustrate experimental probability. • Social justice: the learner plays games where they accept there is winning or losing and acknowledging honest outcomes.				

Pertinent and Contemporary Issues (PCIs):

- Social Cohesion: the learner interacts with others freely as they discuss, create and play games related to probability.
- Life skills and values education: the learner uses different experimental events and lists down all the possible outcomes which assist in decision making.
- Road safety: the learner discusses with peers the probability of getting pedestrians crossing the roads at undesignated areas and highlighting its dangers for public awareness.

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Suggested Assessment Rubric: Statistics and Probability

Indicators	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to draw a frequency distribution table for grouped and ungrouped data and determine mean, mode, and median	The learner accurately and systematically draws a frequency distribution table for grouped and ungrouped data and determines mean, mode, and median	The learner accurately draws a frequency distribution table for grouped and ungrouped data and determines mean, mode, and median	The learner needs assistance to accurately draw a frequency distribution table for grouped and ungrouped data or determines mean, mode, and median	The learner partially draws a frequency distribution table for grouped or ungrouped data or determines mean, mode or median

Ability to represent and interpret grouped data using histograms and frequency polygons	The learner systematically and accurately represents and interprets grouped data using histograms and frequency polygons	The learner accurately represents and interprets grouped data using histograms and frequency polygons	The learner accurately represents or interprets grouped data using histograms or frequency polygons	The learner partially represents or interprets grouped data using histograms or frequency polygons
Ability to perform experiments involving probabilities and identify the range of probability	The learner systematically and accurately performs experiments involving probabilities and identifies the range of probability	The learner accurately performs experiments involving probabilities and identifies the range of probability	The learner accurately performs experiments involving probabilities or identifies the range of probability	The learner partially performs experiments involving probabilities or identifies the range of probability

Indicators	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
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Ability to generate probability space of different events and determine the probability of mutually exclusive and independent events	The learner systematically and accurately generates probability space of different events and determines the probability of mutually exclusive independent events	The learner accurately generates probability space of different events and determines the probability of mutually exclusive independent events	The learner accurately generates the probability space of different events or determines the probability of mutually exclusive or independent events	The learner partially generates probability space of different events or determines the probability of mutually exclusive or independent events
Ability to apply the laws of probability and determine the probability of independent events using tree diagrams	The learner precisely and correctly applies the laws of probability and determines the probability of independent events using tree diagrams	The learner precisely and correctly applies the laws of probability and determines the probability of independent events using tree diagrams	The learner correctly applies the laws of probability or determines the probability of independent events using tree diagrams	The learner partially applies the laws of probability or determines the probability of independent events using tree diagrams

APPENDIX 1: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Strand	Sub-strand	Suggested Assessment Methods	Suggested Learning Resources	Suggested Non-Formal Activities
Numbers and Algebra	Real Numbers	• Class activities • Class written tests	Games on charts, number cards.	Clubs
	Indices and Logarithms	• Class activities • Class written tests	Mathematical tables Calculators	Clubs
	Quadratic Expressions and Equations	• Class activities • Class written tests	Rulers, drawing tools, graph papers/squared books	Clubs
Measurement and geometry	Similarity and Enlargement	• Class activities • Class written tests	Rulers, drawing tools, graph papers/ squared books, geometrical set. Models of similar figures and shapes	Clubs
	Reflection and Congruence	• Class activities • Class written tests	Rulers, drawing tools, graph papers/ squared books, geometrical set.	Clubs

	Rotation	● Class activities ● Class written tests	Rulers, drawing tools, graph papers/ squared books, geometrical set.	Clubs
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Strand	Sub-strand	Suggested Assessment Methods	Suggested Learning Resources	Suggested Non-Formal Activities
Measurement	Trigonometry	● Class activities ● Class written tests	Rulers, drawing tools, graph papers/ squared books, geometrical set, calculators	Clubs
	Area of polygons	● Class activities ● Class written tests	Rulers, drawing tools, geometrical set, calculators, objects with shapes of different polygons, triangular cut outs	Clubs

	Area of a Part of a Circle	• Class activities • Class written tests	Rulers, drawing tools, geometrical set, calculators, objects with shapes of different polygons, circular cut outs	Clubs
	Surface Area and Volume of Solids	• Class written tests • Home or extended Project	Cylinders, cubes, cuboids, cones, spheres, different prisms	Clubs
	Vectors	• Class activities • Class written tests	Rulers, drawing tools, graph papers/ squared books, geometrical set, calculators	Clubs

Strand	Sub-strand	Suggested Assessment Methods	Suggested Learning Resources	Suggested Non-Formal Activities
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	Linear Motion	● Class activities ● Class written tests	Stop watches, rulers/tape measures, rulers, drawing tools, graph papers/squared books, geometrical set, calculators	
Statistics and Probability	Statistics	● Class activities ● Class written tests	Data from different sources, rulers, drawing tools, graph papers/squared books, geometrical set, calculators	
	Probability	● Class activities ● Class written tests	Coins, dice, data from different sources	

APPENDIX 2: USE OF ICT TOOLS

The following ICT tools may be used in learning and teaching of Mathematics at this level:

1. Learner digital devices (**LDD**)
2. Teacher digital devices (**TDD**)
3. Mobile phones
4. Digital clocks
5. Television sets
6. Videos
7. Cameras

8. Projectors
9. Radios
10. DVD players
11. CDs
12. Scanners
13. Internet
14. Other resources.